Faria Mabood

TelecomSystem Engineer| Distributed C++ Architect| AI Driven Automation Engineer
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Professional Summary

Telecom Systems Engineer with 3+ years of experience building carrier-grade distributed C++ systems for OTN/DWDM cross-connect platforms (4.8T / 8T / 12T). Specialised in multi-threaded low-latency architectures, hardware–software integration, restoration orchestration, and AI-driven fault automation. Strong expertise in Linux internals, performance profiling, scalable backend services, and intelligent diagnostic systems for telecom infrastructure.

Technical Skills

Core Systems: C++, Multithreading, Memory Management, Linux Internals

Networking: TCP/IP, Socket Programming, IPC, Distributed Systems

Databases: MySQL (indexing, query optimisation), PostgreSQL

Backend: Python, FastAPI, REST APIs, SQL

DevOps: Docker, Jenkins, Git, CI/CD

AI & Automation: LLMs, Vector Search, PyTorch, AI-based Log Analysis

Professional Experience

Software Developer

Centre for Development of Telematics (C-DOT) – Govt. Of India | Jan 2023 – Present

- Designed and implemented a carrier-grade multi-threaded C++ configuration engine for OTN/DWDM cross-connect systems (4.8T / 8T / 12T platforms).
- Designed and integrated hardware–software communication modules for controller, fabric, and line cards, ensuring seamless real-time device configuration and monitoring.
- Developed optimised MySQL API layer for DWDM/OTN configuration orchestration, improving query performance and real-time system responsiveness.
- Implemented restoration and fault-handling logic for distributed nodes, resolving configuration mismatches, port mapping inconsistencies, and hardware connectivity failures.
- Engineered LLM-powered fault diagnosis system integrating alarm correlation, log analysis, and configuration validation, reducing manual troubleshooting time by 40%.
- Built simulation and validation frameworks for configuration exchange and traffic provisioning, reducing production errors by 30%.
- Optimised multi-threaded C++ modules using GDB and Valgrind profiling, improving memory efficiency and execution speed in real-time environments.
- Contributed to CI/CD deployments, peer code reviews, and Agile sprint planning to ensure high-quality software delivery.

Post Graduate Engineer Trainee | Larsen & Toubro Technology Services | 01/2022 - 07/2022, Mysore

- Developed Python-based REST API for bird species identification using deep learning audio models.
- Built a currency controller Web API that generates dynamic exchange rate tables across regions.

AI/ML Projects

LLM-Based Telecom Fault Diagnosis System

- Designed an AI-driven diagnostic engine for automated root cause analysis of network alarms.
- Built a real-time log ingestion and processing pipeline integrated with telecom management systems.
- Implemented structured reasoning workflow combining alarm correlation, configuration validation, and traffic analysis.
- Achieved 40% reduction in manual troubleshooting effort through automated RCA generation.

Automatic Animal Recognition and Identification | *Deep Learning, Object Detection*

- Implemented a **three-layer approach** comprising body bounding box annotation, keypoint detection, and precise prediction of animal class and bounding box.

Education

Master of Technology (M.Tech) in Computer Science and Engineering

Graduated with **First Division (Distinction) CGPA: 9.42**.

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Graduated with ***First Division CGPA: 8.11.***

Certifications

- Machine Learning - Stanford University(Coursera).
- Getting Started with AWS Machine Learning(Coursera).